

**RAJIV GANDHI COLLEGE OF ENGINEERING AND
TECHNOLOGY**

COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT - 5

Prepared by: Mohamed .B. Sirajuddeen

Reg No : 22TD0053

Subject Name : Enterprise Solutions

Subject Code : CST 61

Year/Sem/Sec : 43 / SG / B

Date of Submission: 21/4/2025

Unit : 5

Teacher's Sign :

Department Vision And Missions

Vision

Cultivate student as technocrats, researchers and entrepreneurs in the field of computer science and engineering.

Mission

- To impart quality education in Computer Science and Engineering by means of learning techniques and value-added courses.
- To inculcate professional excellence and research culture by encouraging projects in cutting-edge technologies through industry interactions.
- To build leadership skills, ethical values and teamwork among the students.
- To strengthen the collaboration of department and industry through internships, mentorships and professional body activities.

1) Describe Business components in detail.

A business component is a logical entity that associates columns from one or more tables into a single structure. Business components provide a layer of wrapping over tables. This creates convenience, developer-friendly naming and separation of development database administration role.

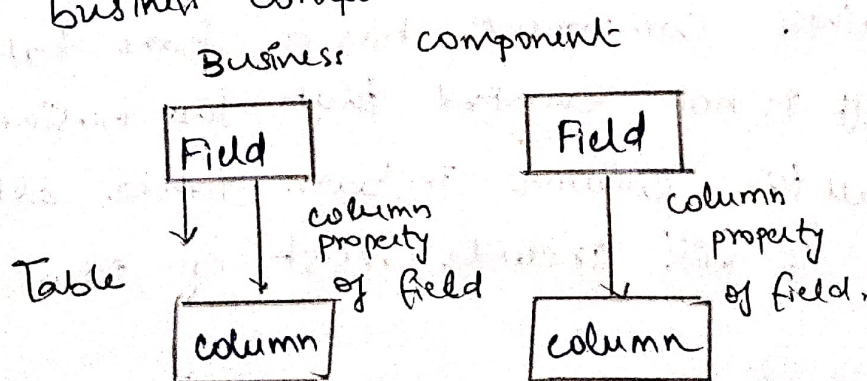
A business component can also have a default sort or search specification, providing records to applets in a predetermined sort order criterion.

When instantiated in a Siebel Application, a business component is comparable to a record set.

Records in a List Applet:

- * A business contains fields. Each column whose data is included in a business component is represented with a field.
- * The field object type is a child of Business component object type.
- * The column object is a child of the Table object type.

The set of field child object definitions of a business component maps a corresponding set of column into the business component.



Component Field and column relationship:

In addition to connecting a column to the business component, the field object type also allows you to specify a meaningful name

that refers to that column. columns in tables are often named in DBMS.

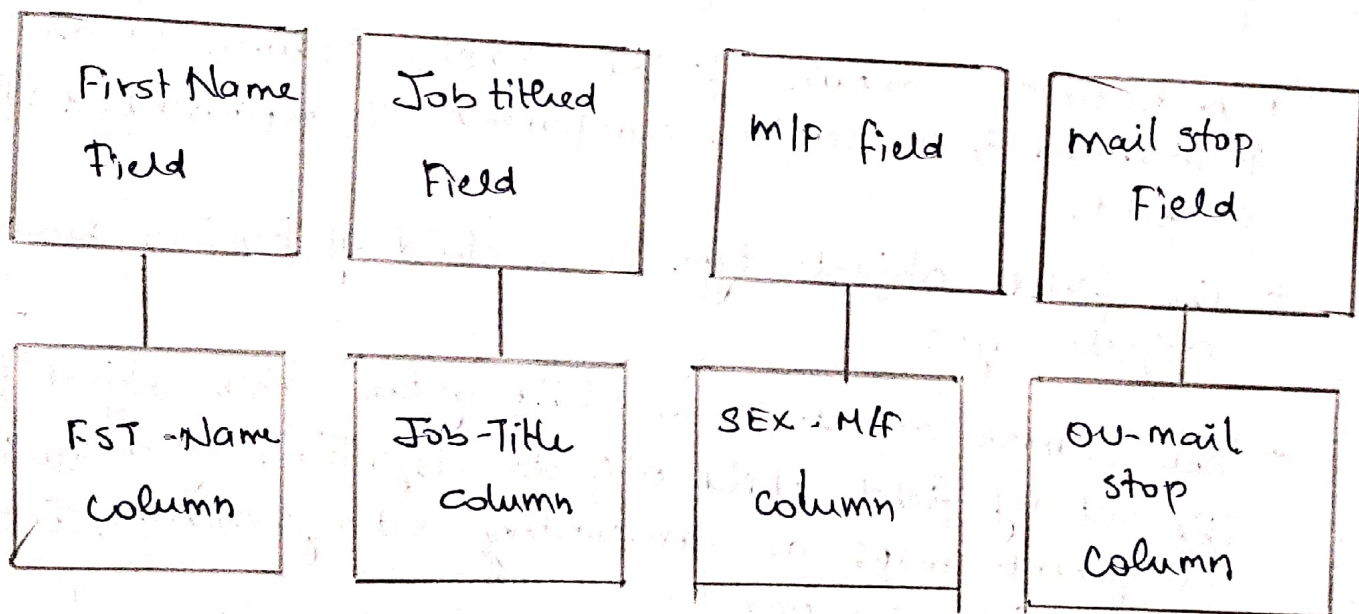
Base Table of business components:

A base Table of a business component is assigned to the business component to provide the most important columns for use as fields in business component.

The base table is assigned to business component with table property in the business component's object definition. The below an example of some field in table 3-CONTACT.

Business component - contact

Base Table - 3-CONTACT



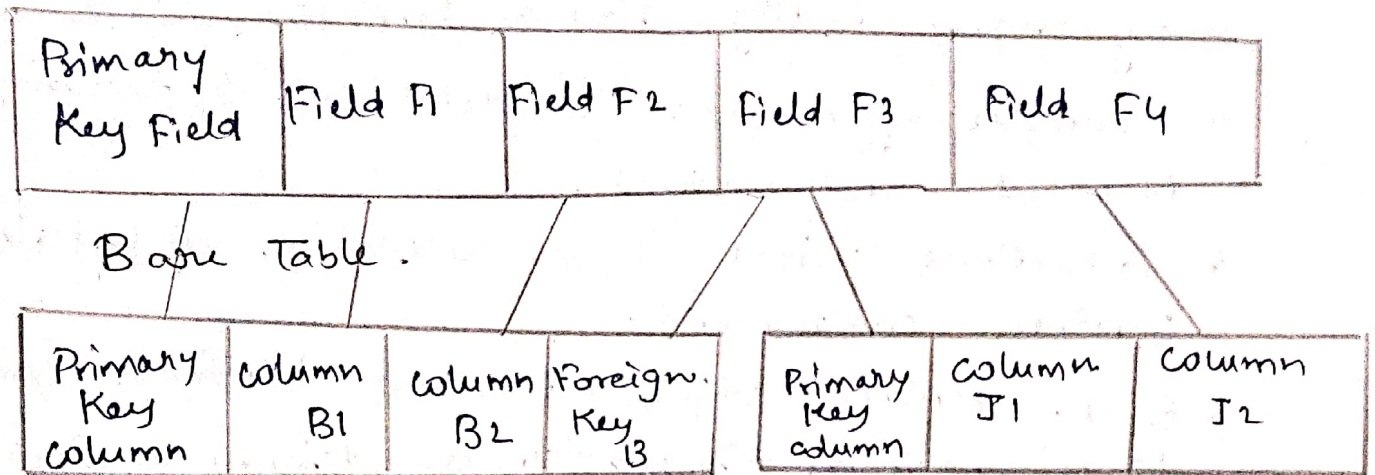
Every business component has a base table assigned to it. It is not evented that the business component include all the columns in base table although typically it will include most of them.

Joined Tables:

A joined tables provides rows on a pre-to one basis to the business component as a result of a foreign key relationship between the joined tables and the business component's base table.

However, not every record in the base table will have joined.

Business component



Extension Tables:

- * There are a special kind of Joined Tables. Like other joined tables, extension's tables provide rows on a one to one basis in parallel within base table rows.
 - * Extension tables are provided specifically to allow column to be virtually added to a ^{base} table rather than physically added.
- Extension Table data, unlike the data in other joined tables can be updated in business component.

Intersection Business components:

An intersection business component is a business component based on an intersection table.

It provides the means to display all of the combinations of data in a many to many relationship, instead of only one or the other one-to-many relationship of which it is composed.

2) Explain business objects in detail.

+ Business objects A business object represent a major functional area of the enterprise - every major entity has a business object.

+ Examples of business object are opportunity, Account and contact.

+ A business object is a collection of related business components.

+ Each business object has one business component that serves as the master or driving business components.

* This focus for business object, have the same name (the name is opportunity)

Business object Types;

Business component:

A business component is logical entity that associates column from one or more tables, causing applets to reference business components rather than the underlying tables.

Data changes made by any one user are reflected in all instances of the business component.

FIELD: A field is ~~also~~ object definition association of a column to a business component. This is now columns from tables are assigned to a business component and provided with meaningful names that the customer developer can easily change.

Business objects:

A business object component object definition is used to include a business component is related to another business component in content of same business.

LINK: A link is a one to many relationship between business components. The link object type make master-detailed views possible.

Multivalue-link:

A multi value link is used in the implementation of a multi value group. A multi-value group is a user maintainable. Find of detail records when it is displayed in a list or form applet.

Join: A join is object definition creates a relationship between a business component and a table field using column from the non-base (Joined) table.

Join specification: A join specification is a child object type of join that provides details about how the join is implemented with the business component.

User property: A user property is a temporary storage filled in a business component that is not tied to the database.

Business service:

A business service is a reusable module containing a set of methods. It provides the ability to call its C++ or script methods from customer-defined scripts and object interface logic mechanism.

3) Explain Siebel E-business Application.

Siebel E-business Application is an integrated suite of customer relationship management (CRM) software developed by Siebel Systems. It is designed to manage customer-facing business functions like

- * Sales
- * Marketing
- * Customer service
- * Call centre

It integrates closely with ERP systems like Oracle E-business suite, SAP and Microsoft Dynamics to deliver a full end-to-end business process solution.

Role of Siebel.

How is it different from ERP

- * ERP = back-office (Finance, Logistics, Inventory)
- * Siebel CRM = front-office (Customer engagement, Lead generation, Issue resolution)

They work together in large organisations to support end-to-end business processes.

Business Modules.

a) Siebel Sales:

- * Handles opportunity lifecycle: Lead → Quote → Order.

- * Integrates with ERP for product master sync, Discount / Approval rules and real time credit check.

b) Siebel Service:

- * Tracks Service Requests, complaints, field visits
- * Interfaces with ERP for: service contract, spare part inventory and billing for time / material.

c) Siebel Field Service.

- * Used by Technicians in real-time via mobile apps.
- * Sync with ERP for parts Availability, service orders and job costing and invoicing.

d) Partner Relationship management.

- * Onboards and supports resellers.
- * ERP shares data like commission calculations, sales volume Tracking and Inventory Allotment.

Components of Siebel Application.

- 1) Siebel Web server extension (SWSE)
- 2) Siebel Application Server
- 3) Siebel Gateway server
- 4) Siebel Database
- 5) Siebel Tools.

Integration with ERP systems.

Siebel uses various methods to integrate with ERP system.

- * Web servers (SOAP / REST)

→ Real Time Integration for data exchange

- * Middleware (eg: Oracle SOA Suite, Tibco)

→ Facilitates reliable data flow and orchestration between systems.

* EAI (Enterprise Application Integration)

→ Siebel EAI Tools support XML, MQSeries, HTTP and JMS.

→ Siebel EIM (Enterprise Integration manager)

→ Used for batch data imports, exports from ERP system.

physical user Interface
object layer

Physical UI file
Based Metadata

①

User Interface object
layer

User Interface
Object Definitions

②

Business objects Layer

Business Object
Definitions

③

Data object Layer

Data Object
Definitions.

④

Siebel
Application

RDBMS



Oracle or Third
party database

- * Unified customer view: combines CRM interaction history with ERP transaction.
- * Faster order processing: Sales & order become orders without duplication.
- * Improved Forecasting: Sales Data Feeds ERP planning modules.
- * Enhanced customer service: Real-Time info on orders, warranties, and billing.
- * Automation: Reduced manual steps in sales, marketing and services.

Real-world enterprises use case.

- * Sales Team uses Siebel for customer onboarding and plan customization.
- * Customer service handles complaints via Siebel call center.
- * ERP system Handles: Inventory provisioning, Invoicing and payments and contract management.
- * Data Flow in real-time between both systems to ensure accurate bills, fast issue resolution.